



Response to Request for Information: Accelerating Speed to Power/Winning the Artificial Intelligence Race: Federal Action to Rapidly Expand Grid Capacity and Enable Electricity Demand Growth

Submitted by the Software & Information Industry Association to the Grid Deployment Office, Department of Energy

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Introduction

The Software & Information Industry Association (SIIA) appreciates the opportunity to respond to the Department of Energy's (DOE) Request for Information on "Accelerating Speed to Power / Winning the Artificial Intelligence Race: Federal Action to Rapidly Expand Grid Capacity and Enable Electricity Demand Growth."

SIIA's nearly 400 members, including creators of software and platforms used by millions worldwide, are at the forefront of artificial intelligence innovation. Our member companies range from global leaders in AI models and applications, companies specializing in data analytics and information services, academic and scientific publishers, education technology companies, and the global financial information and market data community.

Reliable, affordable, and scalable access to electricity is fundamental to the operation and expansion of AI and other emerging technologies. Data centers and AI compute clusters require consistent, high-quality power to support workloads that are central to U.S. economic growth, scientific discovery, and national security. Companies are investing large sums of capital in the United States to expand digital infrastructure, but that infrastructure is only viable if it can connect to a robust, reliable, and reasonably priced electricity grid. However, speed to power remains a key concern. While power lines can often be physically constructed in two to three years, the associated studies, permitting, and siting processes can extend beyond a decade. Over the past 25 years, the United States has not experienced structural electricity demand growth at the scale emerging today, and the systems and supply chains needed to support this expansion are being tested. Capturing the AI opportunity in the United States will require both significant investment and meaningful permitting and planning reforms.

Across the country, we are already experiencing significant and growing power bottlenecks that directly affect data center deployment. In grids without sufficient headroom, new large loads can trigger cascading upgrades that quickly expand out of control due to the interconnected nature of the system. This reality underscores the need for long-term regional transmission lines and backbone infrastructure that can be planned and built in



advance of load, rather than expanding the grid on an ad hoc, project-by-project basis. Because transmission exhibits strong economies of scale, larger backbones built once are far more cost-effective than incrementally rebuilding the grid for each new load or generator that seeks to interconnect.

SIIA appreciates the Trump Administration's sustained focus on addressing energy infrastructure and AI in order to bolster American leadership.¹ These initiatives are essential steps toward aligning U.S. energy infrastructure with the demands of a rapidly evolving digital economy. However, substantial gaps still need to be addressed to support the scale of investment required to support the nation's emerging AI and data infrastructure needs.² We appreciate that DOE is considering actions to strengthen the federal role in accelerating critical generation and transmission projects to ensure the electric grid can accommodate and enable this growth. The comments below elaborate on the challenges created by bottlenecks in interconnection, transmission, and permitting. We offer specific recommendations to improve speed to power, support private investment, and strengthen U.S. competitiveness in AI.

Section 3: Use of DOE Funding, Financing, and Technical Assistance

a. In what specific ways can DOE support the development and deployment of large-scale generation and transmission projects? Please provide concrete examples and suggestions in areas such as: i. Financial incentives (e.g., grants, loans, tax credits, etc.) iv. Streamlining environmental review and permitting processes

The United States currently lacks a cohesive national framework to ensure sufficient and reliable energy supply for data centers and AI compute facilities. Permitting and transmission processes are often fragmented across federal, state, and local jurisdictions, resulting in significant delays and uncertainty for infrastructure projects critical to AI development. State-level policies can further exacerbate these challenges through inconsistent rate structures or discriminatory practices targeting high-load users such as data centers. Additionally, the absence of forward-looking federal incentives for low-carbon, high-reliability energy sources, particularly nuclear, limits the country's ability to scale AI infrastructure sustainably. SIIA member companies rely on large-scale, high-performance computing infrastructure to train and deploy AI models, manage data operations, and deliver digital services globally. These facilities are energy-intensive, and uncertainty around power availability, costs, and permitting timelines directly affects their ability to plan, invest, and expand within the United States. Without reliable access to affordable and sustainable energy, U.S.-based technology firms face increased costs and risk falling behind international

¹ Software & Information Industry Association, [SIIA Commends White House on the AI Action Plan](#) (July 23, 2025).

² Bethany Abbate and Paul Lekas, [America's Energy Grid Needs to Be Ready to Power U.S. Innovation](#) *RealClearEnergy.org* (October 30, 2025).



counterparts operating in jurisdictions with more favorable infrastructure and regulatory environments.

Lengthy or unpredictable permitting timelines also deter private investment, delay innovation, and can push critical AI infrastructure projects offshore, undermining U.S. economic and national security interests. Furthermore, disparate state policies and uneven incentive structures create regional inequities that distort market competition and limit the geographic diversity of AI infrastructure development.

Recommendation: Streamline Permitting for AI Infrastructure Projects

SIIA encourages efforts to streamline federal permitting reforms that simplify and accelerate approvals for data centers, high-performance compute clusters, and related facilities. Such reforms would reduce uncertainty, attract private investment, and help distribute AI infrastructure equitably across regions.

Recommendation: Incentivize Domestic Nuclear Energy for AI Infrastructure

To ensure a stable, low-carbon energy supply capable of meeting AI's increasing power needs, the federal government should incentivize the development of domestic nuclear energy, including small modular reactors (SMRs). This can be achieved through streamlined regulatory pathways, targeted tax incentives, and federal loan guarantees. Investing in next-generation nuclear energy would provide a dependable and sustainable source of power for data centers while advancing U.S. energy security goals.

g. What additional coordination is needed between DOE and other federal agencies (e.g., U.S. Department of Agriculture, Department of Commerce, Department of Interior, Environmental Protection Agency, the Department of Defense, Department of the Treasury, Department of Transportation, the Federal Energy Regulatory Commission, etc.) to align funding, permitting, or policy with emerging electric load challenges?

Recommendation: Strengthen Federal Authority Over Transmission Development

SIIA supports granting greater authority to the Federal Energy Regulatory Commission (FERC) to site and permit interstate transmission lines deemed to be in the national interest. Enhanced federal coordination would streamline the approval process for energy infrastructure, reduce permitting delays, and ensure a reliable national grid capable of meeting growing AI energy demands. Where appropriate, the federal government should have the ability to utilize eminent domain to advance projects critical to national competitiveness and energy security.

Recommendation: Prevent Discriminatory State-Level Practices



The federal government should implement policies or guidance to prevent states from enacting discriminatory energy pricing or regulatory barriers that disproportionately burden data centers or innovations in AI-related infrastructure. Establishing a national standard for data center electricity rates and ensuring equal access to economic incentives across jurisdictions would foster a level playing field for AI infrastructure investment. This would also help avoid regional concentration of compute resources and promote more balanced economic development in communities across the country as America increasingly harnesses the benefits of AI.

Section 4: Load Growth Trends

a. What types of new electric load are driving demand increases in your service area or region? b. Please provide any available projections or forecasts of the scale, timing, and location of this expected growth.

The expansion of AI in the United States has created a real and sustained increase in energy demand for digital infrastructure and cloud computing. While numerous large-scale projections have been produced to quantify this growth, they must be interpreted with extreme caution. The methodologies currently used to forecast AI-related load growth are often opaque, inconsistent, and may not yet be fit for purpose in an era of rapid digital expansion.

The scale of projected growth is substantial. For example, based on data submitted by utilities, PJM is now expecting well over 60,000 MW of new data center demand by 2030. However, these projections illustrate both the magnitude of potential load and the deficiencies of current forecasting practices. Many forecasts are built on an aggregation of speculative interconnection requests rather than firm, commercially committed projects. Developer “shopping” exacerbates this problem: prospective developers may submit interconnection requests for the same potential load at multiple utilities in order to secure the most favorable terms. Since these speculative requests are not rigorously vetted for commercial viability, they can appear multiple times in load forecasts, leading to significant double-counting and massively inflated demand projections.

Additionally, there is no consistent standard across utilities and regions. Some utilities include 100 percent of a project’s nameplate capacity in their forecasts as soon as a study agreement is signed. Others apply non-standard ramp rates or utilization factors that are not transparent to stakeholders. This “black box” approach creates extreme unreliability. At the same time, while aggregate load growth is both substantial and growing, large-load interconnection queues generally do not distinguish effectively between firm plans and prospective concepts. Each interconnection request can exacerbate capacity shortfalls and overwhelm planning and queue processes, turning otherwise routine interconnection studies into decade-long exercises. Similar to generation interconnection queues, load queues are attempting to model additions in meshed grids with complex interactive effects,



where each project materially influences the impact of others on both the generation and load sides, yet these queues have historically been managed in isolation.

DOE's recent effort to expedite large-load interconnection via processes at FERC is directed at the right underlying problem. However, it is important to recognize that interconnection study delays are often downstream of a more fundamental constraint: a lack of regional transmission headroom. When the grid lacks baseline capacity, system impact studies become more complex, triggering cascading upgrades that would be better addressed through long-term regional backbone development rather than incremental project-specific fixes.

Recommendation: Base Forecasts on Firm Financial Commitments

To address these challenges, forecasting processes should be reformed to include new large loads only when they are backed by significant up-front investments and firm financial commitments. The current black-box forecasting approach is not a trivial issue. Inflated load forecasts are a primary input to regional capacity markets and can lead to artificially high capacity costs, potentially on the order of tens of billions of dollars per year, that are ultimately borne by all consumers.

This single change would solve two problems:

1. **It protects consumers:** It would ensure that billions of dollars in grid investments are not made to serve "phantom" loads that never materialize.
2. **It enables the AI transition:** The multi-billion-dollar investments in new generation and data center infrastructure required to power the AI revolution depend on stable, accurate, and reliable market price signals. By basing forecasts on economic reality, we create the market confidence necessary for developers of both generation and load to commit capital and build the infrastructure our nation needs.

The inability to uplift and fast-track high commitment loads, either with financial or long-term contractual commitments, is leading to both capacity markets and large load queues. Timely development is only possible when the underlying commitments and processes are clear and dependable.

Section 5: Grid Infrastructure Constraints

a. What generation, transmission, or distribution constraints are limiting the ability to serve this demand? b. What are the primary challenges and barriers to expanding infrastructure and deploying large-scale generation and transmission projects?

SIIA members face a range of generation, transmission, and distribution constraints that limit the ability to serve fast-growing AI and data center loads. At the transmission level, insufficient use of advanced transmission technologies, such as high-capacity advanced



conductors, constrains the grid's ability to quickly expand power flow. Advanced conductors such as aluminum conductor composite core can substantially increase current-carrying capacity, yet many utilities do not allow large-load customers to finance or deploy these upgrades due to technical or process concerns. This limits a practical route for large loads to invest in rapid capacity increases that could meaningfully improve speed to power.

Capacity delays also arise from long interconnection study times and persistent queue backlogs. In many cases, even where storage or co-located generation could be deployed quickly to meet peak needs or provide grid support, regulatory rules make it difficult to use these resources efficiently. There is substantial interest among data centers and other large customers in investing in demand response and hybrid resources such as solar-plus-storage, which can offer additive capacity value for peak shaving and resilience. Yet in many states and utility regions, it is not legally or practically possible to develop capacity resources that directly increase speed to power or to secure appropriate accreditation and compensation for such investments.

On the generation side, nuclear investments are slowed by long lead times for core equipment, extended site diligence requirements, and mandatory hearings that add to timelines even when they are uncontested. More broadly, grid-related lead times are driven not only by transmission overloads but also by delays in procuring and installing equipment needed to stabilize the grid. Large-scale transformers can have multi-year delays, and STATCOMs and other reactive support equipment often face similar or longer timelines. Upgrading lines can trigger the need for this additional equipment, further extending project schedules and complicating planning.

DOE can help address these constraints by using tools such as the Defense Production Act to accelerate American manufacturing of critical grid equipment, including transformers and stability devices. There is an opportunity to leverage targeted investments to pull forward manufacturing capacity and grid buildout, but such efforts require utilities to actively participate and ensure that large-load customers paying a premium for speed are treated fairly with respect to both grid access and the equipment they effectively finance. The fact that this sort of market-based coordination is not occurring at scale today reflects a combination of cultural, supply-chain, and regulatory barriers.

Beyond equipment constraints, broader permitting and siting challenges for energy infrastructure can further stall deployment. The potential for long-term litigation, combined with the absence of upfront clarity and predictable processes, significantly increases project risk. Identifying ways to reduce this risk, whether through new legislation or via expedited and harmonized federal permitting processes and unified siting authorities, will be critical to unlocking timely development of new generation and transmission with sufficient headroom for future growth.



Historically, the United States demonstrated what is possible in the postwar period, when nearly 80,000 miles of transmission were planned, sited, and built within two to five years. By contrast, in the modern environment, major lines can take a decade or more from initial need identification to completion. These kinds of delays will continue to constrain development if left unaddressed. Experience in regions such as Texas shows that where unified siting authority, clear jurisdictional lines, and proactive planning exist, it is possible to expand transmission infrastructure much more rapidly and at lower cost.

Conclusion

The United States stands at a pivotal moment in the global race to develop and deploy artificial intelligence. The ability of U.S. firms to lead in AI depends not only on data, talent, and capital, but also on a power system that can reliably and affordably support large-scale compute. Today's interconnection bottlenecks, forecasting challenges, equipment shortages, and permitting barriers pose risks to that leadership if left unaddressed.

SIIA urges DOE and its federal partners to adopt policies that modernize forecasting methodologies, expand transmission and generation capacity through coordinated planning, and streamline permitting across jurisdictions. Doing so will protect consumers, create clear investment signals, and support the scale of infrastructure necessary to sustain U.S. innovation and competitiveness in AI. SIIA and its member companies stand ready to work with DOE and other federal stakeholders to help design and implement solutions that ensure the United States can meet the energy demands of the AI era.

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